

Write down the definition of the following, include the unit of measurement and the name of the equipment used to measure the following:

Current-

Potential difference-

Investigating I-V Relationships in Circuits

Current is...

‘the flow of charged electrons round a circuit. It is measured in amps, using an ammeter’.

Potential difference is...

‘the difference in electrical potential between two points in a circuit. It is also known as voltage. It is measured in volts, using a voltmeter’.



Required practical- I-V characteristics

Learning Objective:

1. • Understand how an experiment can be designed to test an idea.
2. • Evaluate how an experimental procedure can yield more accurate data.
3. • Interpret and explain graphs using scientific ideas.



Queen
Elizabeth
School

KEYWORDS

Ohms law

Filament lamp

Diode

Resistor

Ohmic conductor

Independent variable

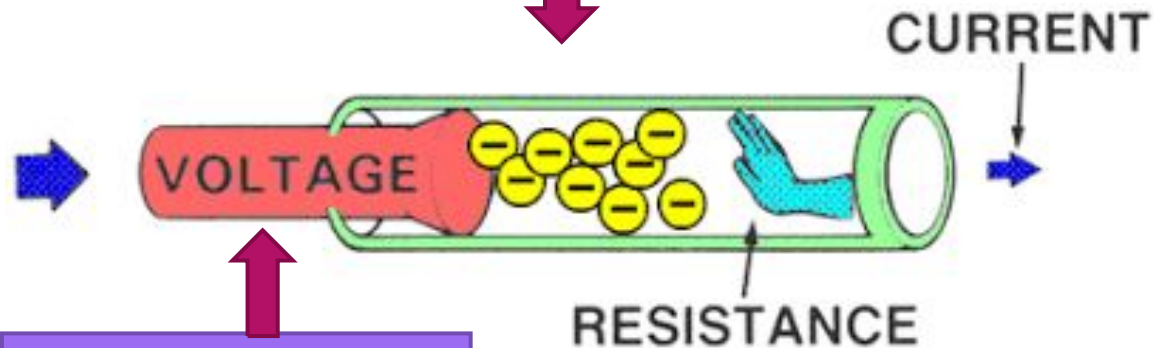
Dependent variable

Background Information

Current:

The flow of electrons

- ▶ What are current, voltage, resistance and difference?
- ▶ How are they linked?

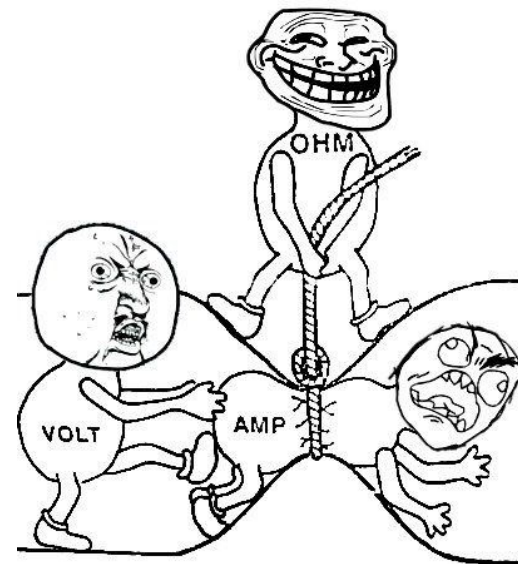


Potential Difference/Voltage:

The amount of 'push' given by the battery to the electrons

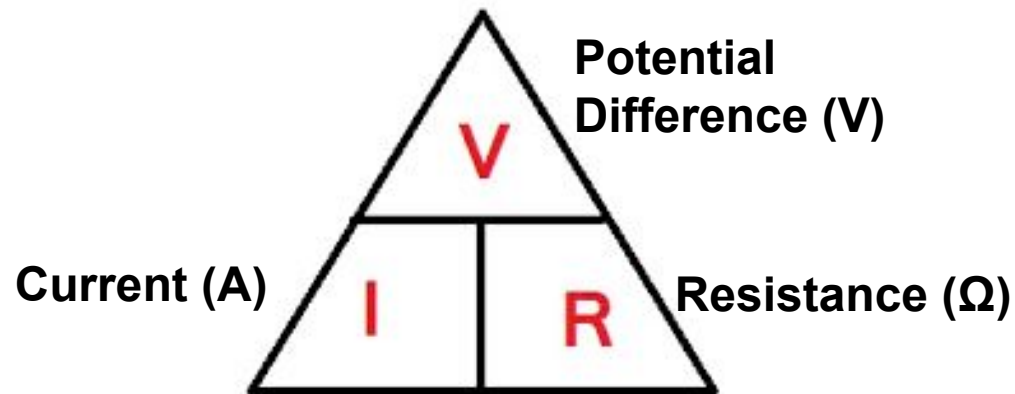
Resistance:

How hard it is for the electrons to get through the wire



Background Information

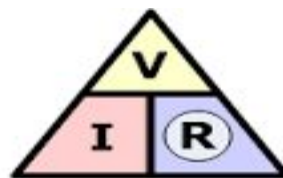
- ▶ What are current, potential difference and resistance?
- ▶ How are they linked?



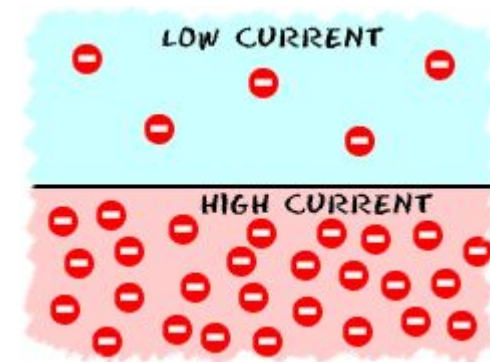
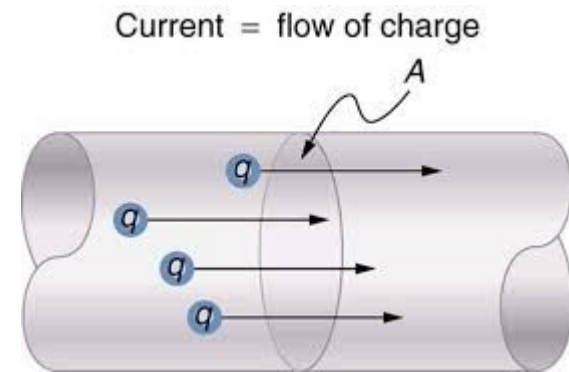
$$V = I \times R$$



$$I = \frac{V}{R}$$



$$R = \frac{V}{I}$$



Investigating I-V Relationships in Circuits

Hypothesis:

The relationship between current, voltage and resistance is defined by Ohm's Law:

$$I = V/R$$

(current = voltage $\div$ resistance)

The relationship between current and voltage are directly proportional.

Ohmic and Non-ohmic components

- Ohmic conductors- are conductors that follow ohm's law. Electric current is proportional to voltage. They have a constant resistance.
-
- Non-ohmic conductors- is a material that does not follow Ohm's law. In a non-ohmic conductor, the relationship between voltage and current is different due to factors like temperature, voltage, and the material of the conductor.
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We can test whether components are ohmic or non-ohmic by testing them, plotting graphs of the data and looking at the shape of the graph.

Prediction

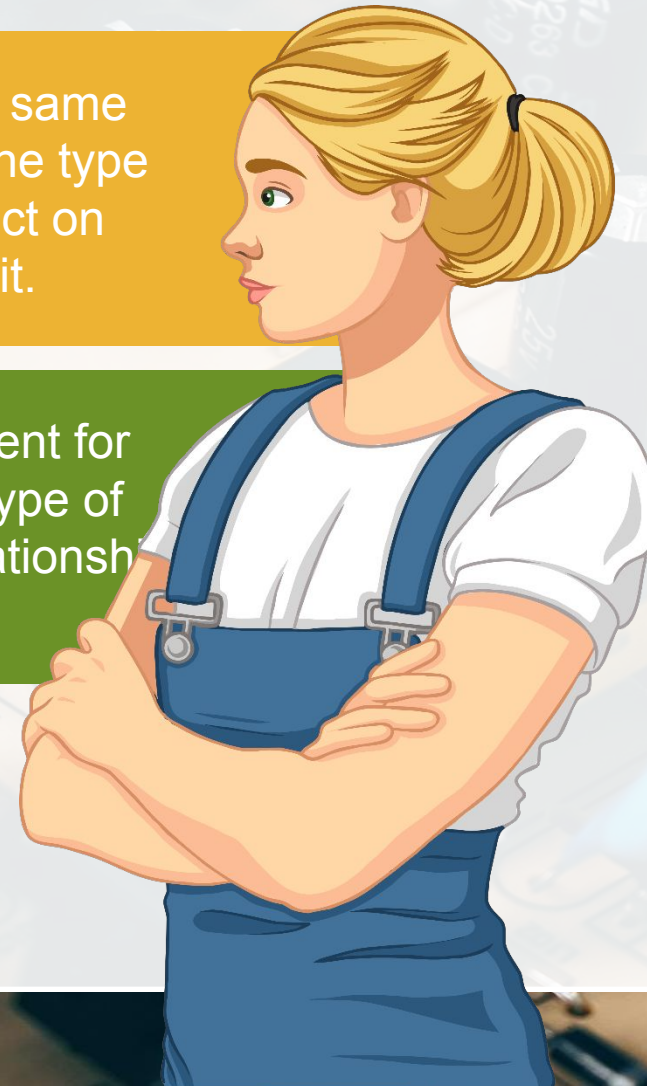


Will the I-V characteristics be the same for all components?

Prediction

The I-V relationship will be the same for all components, therefore the type of component will have no effect on the I-V relationship in the circuit.

The I-V relationship will be different for each component, therefore the type of component will affect the I-V relationship in the circuit.



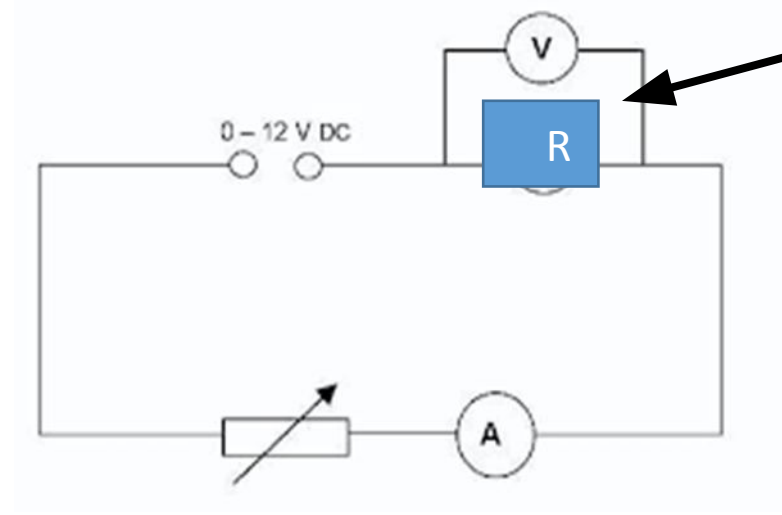
Testing the components: Ohmic resistor, Diode and a filament lamp

- The circuit diagram can be used to check components to see if the current and voltage changes.

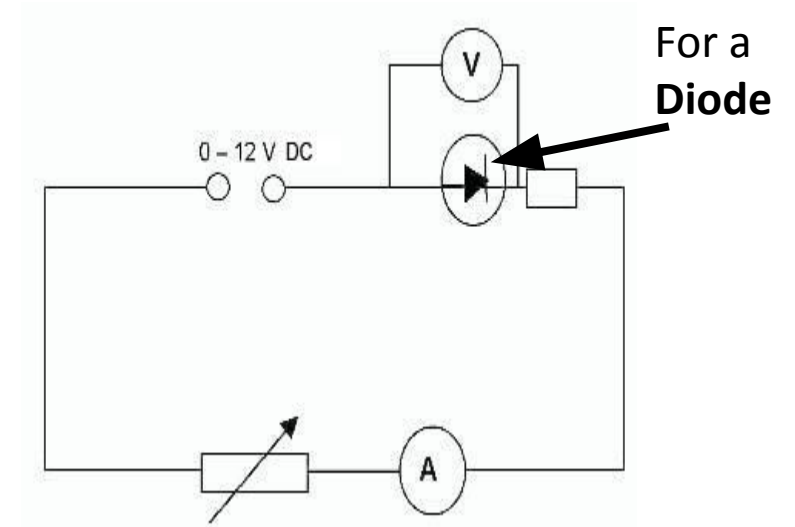
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- The two variables are :
 - Potential difference
 - Current

- <https://www.youtube.com/watch?v=ksPfzUjMbBk>

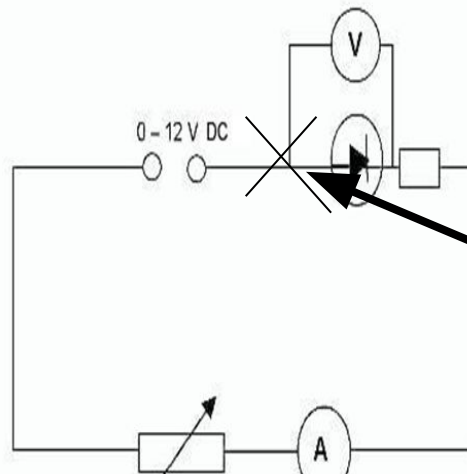
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For an
ohmic
Resistor



For a
Diode



For a
Filament
Lamp

Equipment List

Risk assessment – please read before collecting your equipment.

Ammeter – to measure the current

Voltmeter – to measure the potential difference

Filament Lamp – to observe component characteristics

Ohmic Resistor– to observe component characteristics

Diode – to observe component characteristics

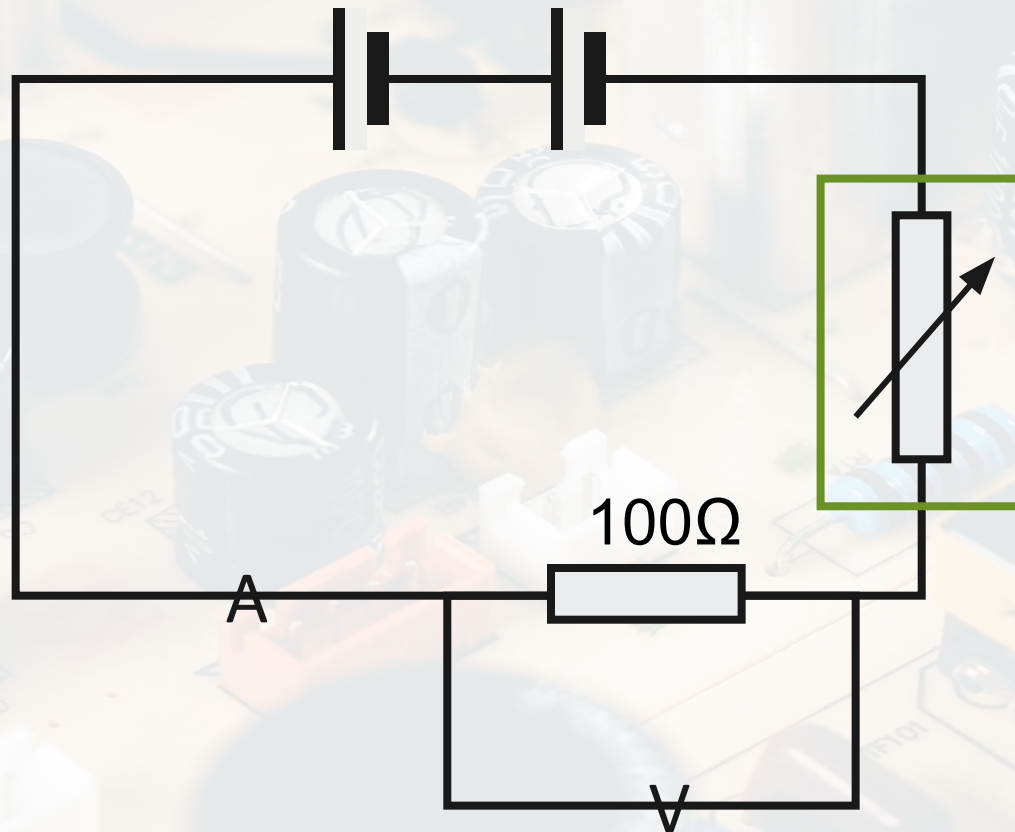
Variable resistor (10Ω) – for use with the diode

Connecting leads – to complete the circuit

Crocodile clips – to connect wired components

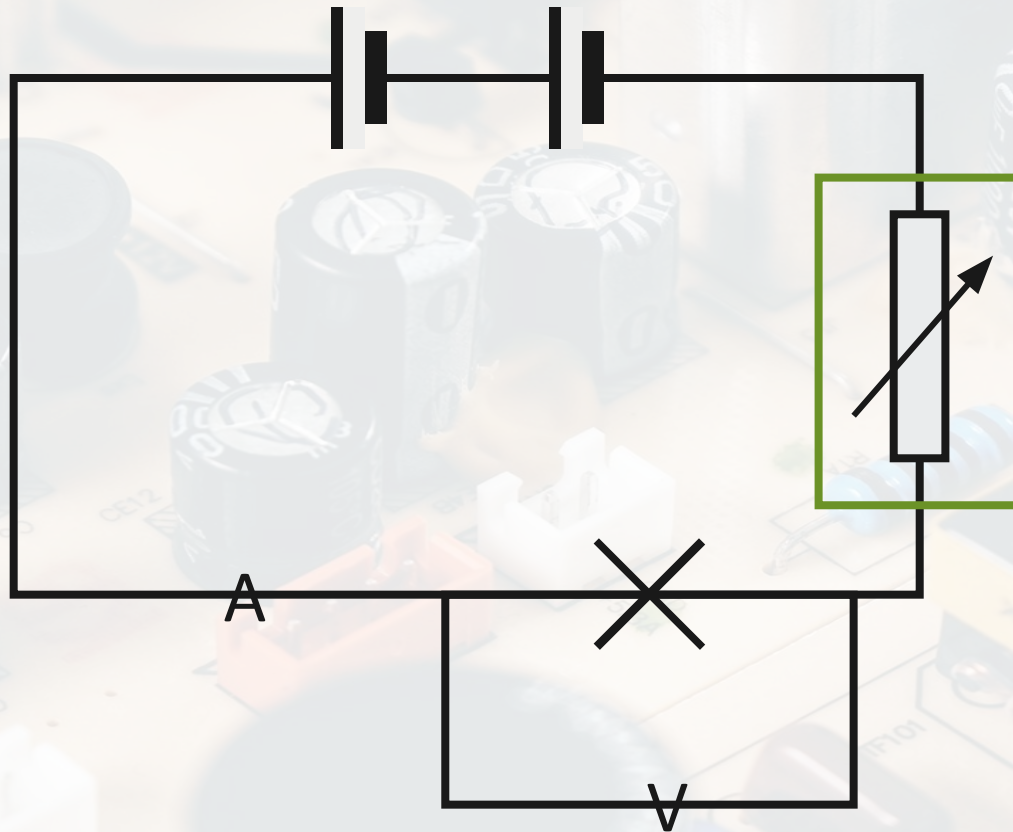
Battery pack – to provide a power supply

Diagram – Resistor



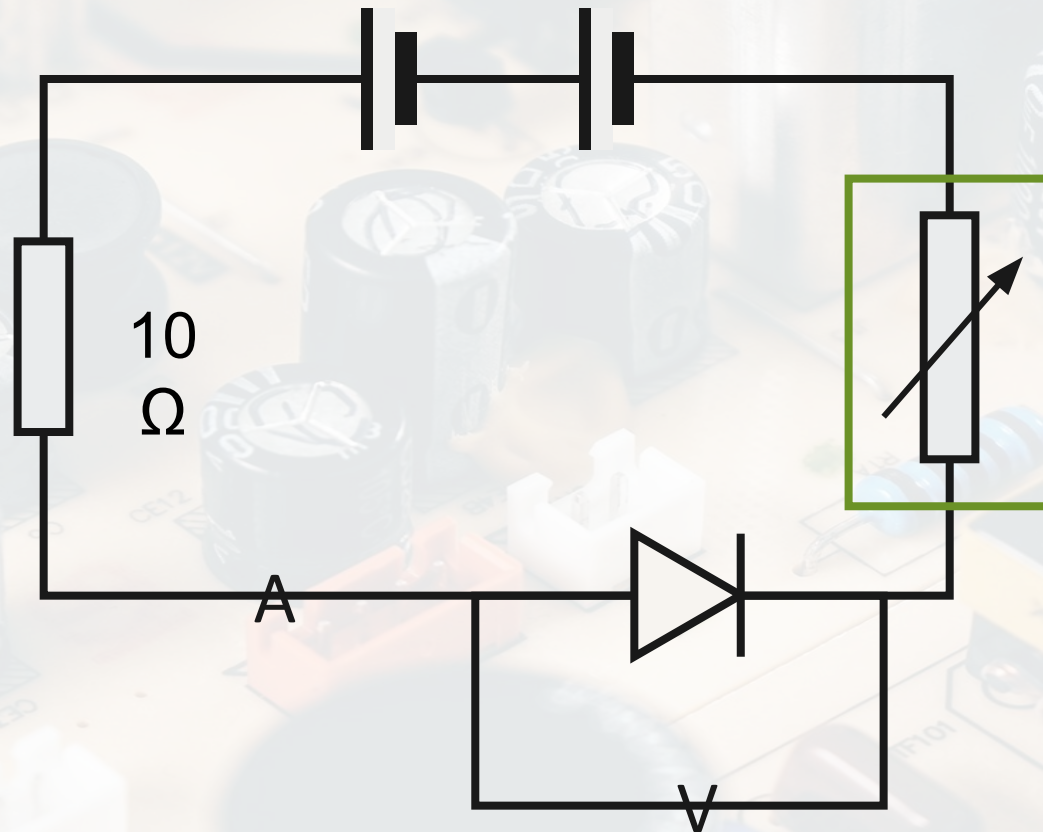
If your lab pack has adjustable voltage settings you will not need to use a variable resistor in your circuit.

Diagram – Filament Lamp



If your lab pack has adjustable voltage settings you will not need to use a variable resistor in your circuit.

Diagram – Diode



If your lab pack has adjustable voltage settings you will not need to use a variable resistor in your circuit.

Results Table

- Current (A)
- Voltage (V)
- Resistance (Ω)



$$\textcircled{V} = I \times R$$



$$\textcircled{I} = \frac{V}{R}$$



$$\textcircled{R} = \frac{V}{I}$$

Remember!

1. Collect three results in one direction THEN change direction of battery and collect three more
2. Complete results table to appropriate degree of accuracy (2 d.p)

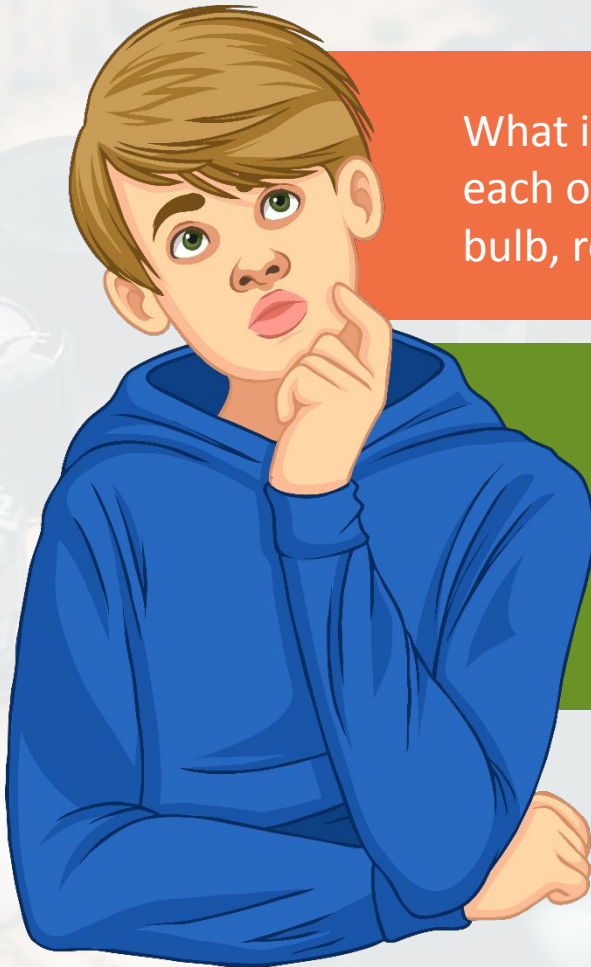
The problems with the above experiment

- The resistance of ohmic components increase when their temperature increases. So as current is passed through the resistor its temperature increases.
- To overcome this problem, the power supply should be turned off between readings during the experiment.

Analysing the data collected

- The readings for pd and current can be plotted as points on a graph.
- A line of best fit can be drawn for each component tested.
- The pd can be applied either way round so negative values can be gathered as well as positive.
- So the axes need to have negative values as well as positive ones.

Conclusion

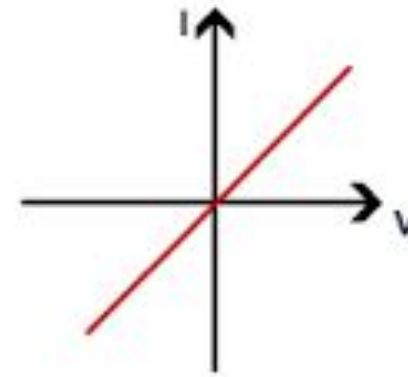


What is the relationship between I-V for each of the three components; filament bulb, resistor and diode?

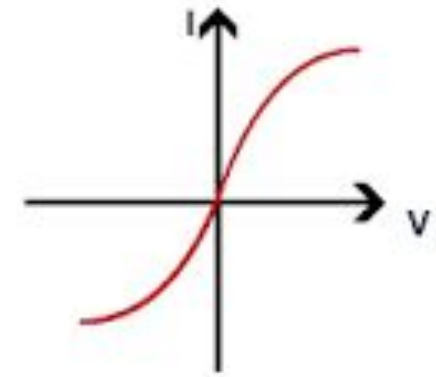
Explain how you used the graph to identify the I-V characteristic of each of the three components; filament bulb, resistor and diode.

Drawing conclusions from the experiment

- By looking at the shape of graphs we can see if the component is ohmic
- If the current is directly proportional to the potential difference then the component is ohmic.

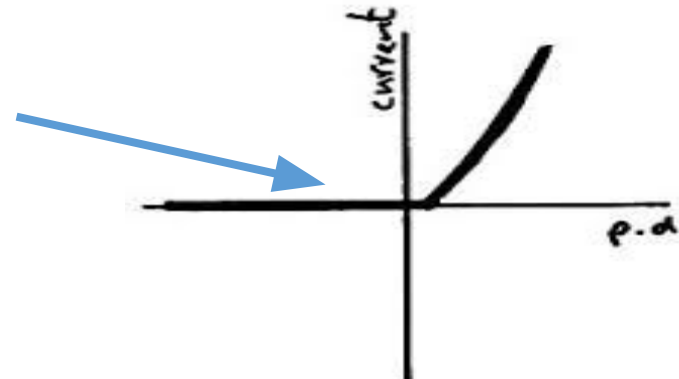


An ohmic resistor.
Wire at constant temperature.



A filament lamp
(non-ohmic).

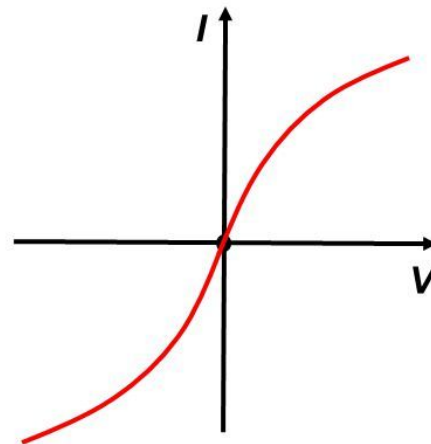
I-V characteristics
for a Diode



Results analysis

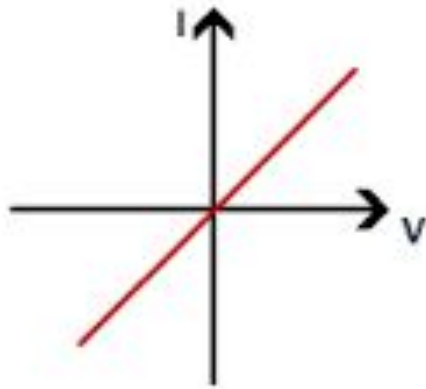
1. What does the graph tell us about resistance across a filament lamp?
2. Why do you think this happens?

Filament lamp

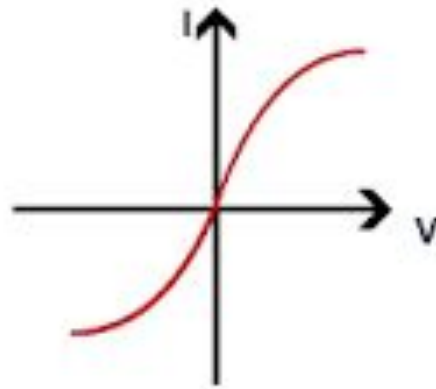


Application

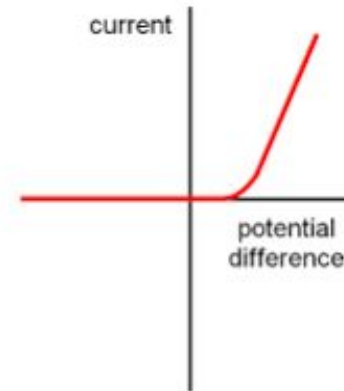
1. What do you notice about the IVR relationship of different components?
2. What might cause this?



An ohmic resistor.
Wire at constant temperature.

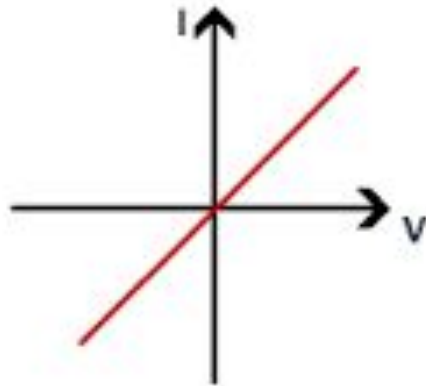


A filament lamp
(non-ohmic).

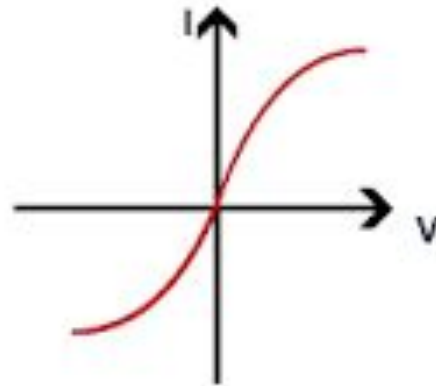


A Diode

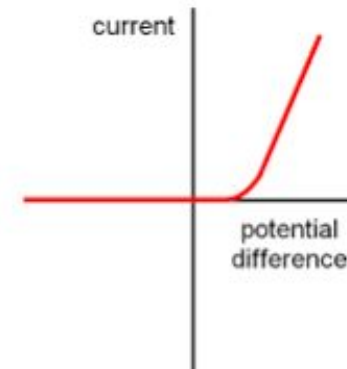
Application



- Ohmic resistor (constant temp)
- I is directly proportional to V
- Straight line

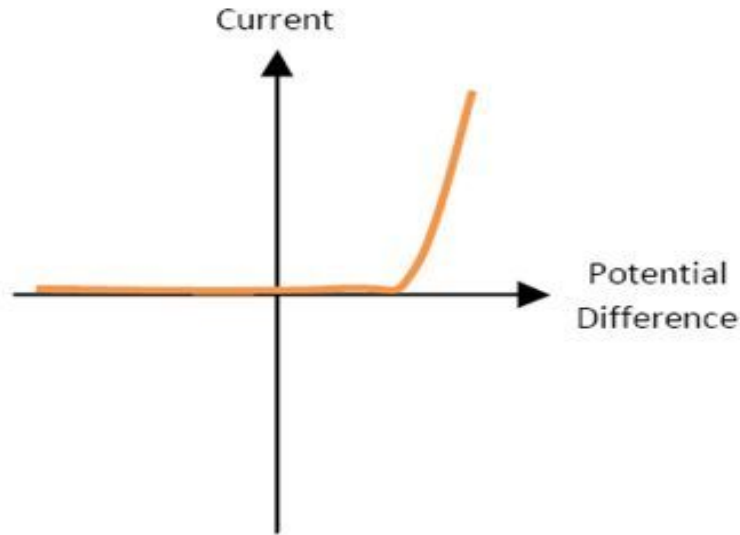


- **Filament Lamp**
- Temperature increases as current increases
- So resistance increases
- Therefore it's harder for current to flow
- And the graph gets less steep



- **Diode**
- Current only flows in one direction
- The diode has a very high resistance in the opposite direction (so no current)

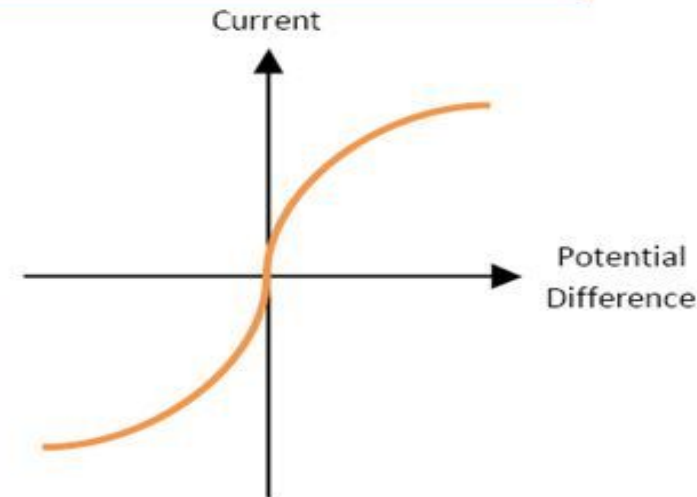
I-V graphs



A diode

Diodes behave like ohmic resistors when the current is travelling through them in the correct direction. However, if the current is reversed the resistance of the diode is extremely high.

The temperature of a filament lamp increase as current increases. Increasing temperature increases resistance.



A filament lamp